2.08 Complex Numbers: C + iS

Worked Solutions

1 a)

Geometric with common ratio $r = \frac{1}{3}$.

1 b)

Not geometric. 5 terms so coefficients are 1, 4, 6, 4, 1:

$$1 + 0.8 + 0.24 + 0.032 + 0.0016$$
$$= 1 \times 1 + 4 \times 0.2 + 6 \times 0.04 + 4 \times 0.008 + 1 \times 0.0016$$
$$= 1 \times 1 + 4 \times 0.2 + 6 \times 0.2^2 + 4 \times 0.2^3 + 1 \times 0.2^4$$
$$= (1 + 0.2)^4.$$

So binomial series.

1 c)

Geometric with common ratio $r = -\frac{1}{4}$.

1 d)

Not geometric. 4 terms so coefficients are 1, 3, 3, 1:

$$8 - 1.2 + 0.06 - 0.001 = 1 \times 8 + 3 \times (-0.4) + 3 \times 0.02 + 1 \times (-0.001)$$
$$= 1 \times 2^3 + 3 \times 2^2 \times (-0.1) + 3 \times 2 \times (-0.1)^2 + 1 \times (-0.1)^3$$
$$= (2 - 0.1)^3.$$

So binomial series.

2 a)

$$a = 4, \qquad r = \frac{1}{3}e^{i\theta}.$$
$$S_\infty = \frac{a}{1-r} = \frac{4}{1 - \frac{1}{3}e^{i\theta}} = \frac{12}{3 - e^{i\theta}}.$$

2 b)

$$a = 1, \qquad r = \frac{1}{4}e^{i\theta}.$$
$$S_\infty = \frac{a}{1-r} = \frac{1}{1 - \frac{1}{4}e^{i\theta}} = \frac{4}{4 - e^{i\theta}}.$$

2 c)

$$a = 2, \qquad r = \frac{1}{5}e^{2i\theta}.$$
$$S_\infty = \frac{a}{1-r} = \frac{2}{1 - \frac{1}{5}e^{2i\theta}} = \frac{10}{5 - e^{2i\theta}}.$$

2 d)

$$a = 5, \qquad r = \frac{1}{10}e^{3i\theta}.$$

$$S_\infty = \frac{a}{1-r} = \frac{5}{1-\frac{1}{10}e^{3i\theta}} = \frac{50}{10-e^{3i\theta}}.$$

2 e)

$$a = 6, \qquad r = \frac{1}{12}e^{-i\theta}.$$

$$S_\infty = \frac{a}{1-r} = \frac{6}{1-\frac{1}{12}e^{-i\theta}} = \frac{72}{12-e^{-i\theta}}.$$

2 f)

$$a = 2, \qquad r = \frac{1}{6}e^{-4i\theta}.$$

$$S_\infty = \frac{a}{1-r} = \frac{2}{1-\frac{1}{6}e^{-4i\theta}} = \frac{12}{6-e^{-4i\theta}}.$$

3 a)

$$a = e^{i\theta}, \quad r = e^{i\theta}, \quad n = 12.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{e^{i\theta}(1-e^{12i\theta})}{1-e^{i\theta}} = \frac{e^{i\theta}-e^{13i\theta}}{1-e^{i\theta}}.$$

3 b)

$$a = 2, \quad r = \frac{1}{2}e^{i\theta}, \quad n = 10.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{2(1-\frac{1}{2^{10}}e^{10i\theta})}{1-\frac{1}{2}e^{i\theta}}$$
$$= \frac{2-\frac{2}{1024}e^{10i\theta}}{1-\frac{1}{2}e^{i\theta}} = \frac{1024-e^{10i\theta}}{512-256e^{i\theta}}.$$

3 c)

$$a = 1, \quad r = -\frac{1}{3}e^{i\theta}, \quad n = 9.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1(1+\frac{1}{3^9}e^{9i\theta})}{1+\frac{1}{3}e^{i\theta}}$$
$$= \frac{1+\frac{1}{19683}e^{9i\theta}}{1+\frac{1}{3}e^{i\theta}} = \frac{19683+e^{9i\theta}}{19683+6561e^{i\theta}}.$$

3 d)

$$a = 16, \quad r = -\frac{1}{4}e^{2i\theta}, \quad n = 8.$$

$$\begin{aligned} S_n &= \frac{a(1-r^n)}{1-r} = \frac{16(1-\frac{1}{4^8}e^{16i\theta})}{1+\frac{1}{4}e^{2i\theta}} \\ &= \frac{16-\frac{1}{4096}e^{16i\theta}}{1+\frac{1}{4}e^{2i\theta}} = \frac{65536-e^{16i\theta}}{4096+1024e^{2i\theta}}. \end{aligned}$$

4 a)

$$z^* = a - ib.$$

4 b)

$$z^* = r(\cos\theta - i\sin\theta) = r(\cos(-\theta) + i\sin(-\theta)).$$

4 c)

$$z^* = re^{-i\theta}.$$

5 a)

$$z^* = 5 + e^{-i\theta}.$$

5 b)

$$z^* = 3 - e^{i\theta}.$$

5 c)

$$z^* = 7 + e^{-2i\theta}.$$

5 d)

$$z^* = 7 + 9e^{-i\theta}.$$

5 e)

$$z^* = 2 - 3e^{-i\theta}.$$

5 f)

$$z^* = 1 + e^{i\theta}.$$

5 g)

$$z^* = 2 - e^{-i\theta}.$$

5 h)

$$z^* = 5 - 5e^{i\theta}.$$

5 i)

$$z^* = 6 - 8e^{7i\theta}.$$

5 j)

$$z^* = 2 + 15e^{-i\theta/2}.$$

6 a)

$$\begin{aligned}(1 + 11e^{-i\theta})(1 + 11e^{i\theta}) &= 1 + 11e^{i\theta} + 11e^{-i\theta} + 121e^{-i\theta}e^{i\theta} \\ &= 122 + 11e^{i\theta} + 11e^{-i\theta}.\end{aligned}$$

6 b)

$$\begin{aligned}(4 + e^{3i\theta})(4 + e^{-3i\theta}) &= 16 + 4e^{3i\theta} + 4e^{-3i\theta} + e^{3i\theta}e^{-3i\theta} \\ &= 17 + 4e^{3i\theta} + 4e^{-3i\theta}.\end{aligned}$$

6 c)

$$\begin{aligned}\left(2 + \frac{1}{2}e^{i\theta}\right)\left(2 + \frac{1}{2}e^{-i\theta}\right) &= 4 + e^{i\theta} + e^{-i\theta} + \frac{1}{4}e^{i\theta}e^{-i\theta} \\ &= \frac{17}{4} + e^{i\theta} + e^{-i\theta}.\end{aligned}$$

7 a)

$$z = \cos\theta + i\sin\theta.$$

7 b)

$$e^{i\theta} = \cos\theta + i\sin\theta \quad \text{and} \quad e^{-i\theta} = \cos\theta - i\sin\theta.$$
$$e^{i\theta} + e^{-i\theta} = \cos\theta + i\sin\theta + \cos\theta - i\sin\theta = 2\cos\theta.$$
$$\therefore \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}.$$

8 a)

$$10 + 6e^{i\theta} + 6e^{-i\theta} = 10 + 6(e^{i\theta} + e^{-i\theta}) = 10 + 6(2\cos\theta) = 10 + 12\cos\theta.$$

8 b)

$$5 - 2e^{4i\theta} - 2e^{-4i\theta} = 5 - 2(e^{4i\theta} + e^{-4i\theta}) = 5 - 2(2\cos 4\theta) = 5 - 4\cos 4\theta.$$

8 c)

$$\begin{aligned}(1 + e^{3i\theta})(1 + e^{-3i\theta}) &= 1 + e^{3i\theta} + e^{-3i\theta} + e^{3i\theta}e^{-3i\theta} \\ &= 1 + (e^{3i\theta} + e^{-3i\theta}) + 1 = 2 + 2\cos 3\theta.\end{aligned}$$

8 d)

$$\begin{aligned}(2 + e^{2i\theta})(2 + e^{-2i\theta}) &= 4 + 2e^{2i\theta} + 2e^{-2i\theta} + e^{2i\theta}e^{-2i\theta} \\ &= 4 + 2(e^{2i\theta} + e^{-2i\theta}) + 1 = 4 + 2(2\cos 2\theta) + 1 = 5 + 4\cos 2\theta.\end{aligned}$$

8 e)

$$\begin{aligned}\left(1 - \frac{1}{2}e^{i\theta/3}\right)\left(1 - \frac{1}{2}e^{-i\theta/3}\right) &= 1 - \frac{1}{2}(e^{i\theta/3} + e^{-i\theta/3}) + \frac{1}{4}e^{i\theta/3}e^{-i\theta/3} \\ &= 1 - \frac{1}{2}(2\cos\tfrac{1}{3}\theta) + \frac{1}{4} = \frac{5}{4} - \cos\tfrac{1}{3}\theta.\end{aligned}$$

9 a)

$$\begin{aligned}\frac{1}{2 - e^{i\theta}} \times \frac{2 - e^{-i\theta}}{2 - e^{-i\theta}} &= \frac{2 - e^{-i\theta}}{4 - 2e^{i\theta} - 2e^{-i\theta} + 1} \\ &= \frac{2 - e^{-i\theta}}{4 - 2(e^{i\theta} + e^{-i\theta}) + 1} = \frac{2 - e^{-i\theta}}{5 - 4\cos\theta}.\end{aligned}$$

9 b)

$$\frac{3}{4+e^{2i\theta}} \times \frac{4+e^{-2i\theta}}{4+e^{-2i\theta}} = \frac{12+3e^{-2i\theta}}{16+4e^{2i\theta}+4e^{-2i\theta}+1}$$
$$= \frac{12+3e^{-2i\theta}}{17+4(e^{2i\theta}+e^{-2i\theta})} = \frac{12+3e^{-2i\theta}}{17+8\cos 2\theta}.$$

9 c)

$$\frac{5}{6-2e^{-i\theta}} \times \frac{6-2e^{i\theta}}{6-2e^{i\theta}} = \frac{30-10e^{i\theta}}{36-12e^{i\theta}-12e^{-i\theta}+4}$$
$$= \frac{30-10e^{i\theta}}{40-12(e^{i\theta}+e^{-i\theta})} = \frac{30-10e^{i\theta}}{40-24\cos\theta} = \frac{15-5e^{i\theta}}{20-12\cos\theta}.$$

9 d)

$$\frac{e^{i\theta}}{1-\frac{1}{3}e^{3i\theta}} \times \frac{1-\frac{1}{3}e^{-3i\theta}}{1-\frac{1}{3}e^{-3i\theta}} = \frac{e^{i\theta}-\frac{1}{3}e^{-2i\theta}}{1-\frac{1}{3}e^{3i\theta}-\frac{1}{3}e^{-3i\theta}+\frac{1}{9}}$$
$$= \frac{e^{i\theta}-\frac{1}{3}e^{-2i\theta}}{\frac{10}{9}-\frac{1}{3}(e^{3i\theta}+e^{-3i\theta})} = \frac{e^{i\theta}-\frac{1}{3}e^{-2i\theta}}{\frac{10}{9}-\frac{2}{3}\cos 3\theta}$$
$$= \frac{9e^{i\theta}-3e^{-2i\theta}}{10-6\cos 3\theta}.$$

9 e)

$$\frac{e^{i\theta}}{1+\frac{1}{8}e^{4i\theta}} \times \frac{1+\frac{1}{8}e^{-4i\theta}}{1+\frac{1}{8}e^{-4i\theta}} = \frac{e^{i\theta}+\frac{1}{8}e^{-3i\theta}}{1+\frac{1}{8}e^{4i\theta}+\frac{1}{8}e^{-4i\theta}+\frac{1}{64}}$$
$$= \frac{e^{i\theta}+\frac{1}{8}e^{-3i\theta}}{\frac{65}{64}+\frac{1}{8}(e^{4i\theta}+e^{-4i\theta})} = \frac{e^{i\theta}+\frac{1}{8}e^{-3i\theta}}{\frac{65}{64}+\frac{2}{8}\cos 4\theta}$$
$$= \frac{64e^{i\theta}+8e^{-3i\theta}}{65+16\cos 4\theta}.$$

10)

$$C+iS = \cos\theta + \frac{1}{4}\cos 3\theta + \frac{1}{16}\cos 5\theta + i(\sin\theta + \frac{1}{4}\sin 3\theta + \frac{1}{16}\sin 5\theta)$$
$$= (\cos\theta + i\sin\theta) + \frac{1}{4}(\cos 3\theta + i\sin 3\theta) + \frac{1}{16}(\cos 5\theta + i\sin 5\theta)$$
$$= e^{i\theta} + \frac{1}{4}e^{3i\theta} + \frac{1}{16}e^{5i\theta}.$$

11)

$$\begin{aligned} C + iS &= 1 - \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos 4\theta + i(\sin\theta - \frac{1}{2}\sin 2\theta + \frac{1}{4}\sin 4\theta) \\ &= 1 - \frac{1}{2}(\cos 2\theta + i\sin 2\theta) + \frac{1}{4}(\cos 4\theta + i\sin 4\theta) \\ &= e^0 - \frac{1}{2}e^{2i\theta} + \frac{1}{4}e^{4i\theta}. \end{aligned}$$

12 a)

$$\begin{aligned} \left(1 + \frac{1}{3}e^{3i\theta}\right)\left(1 + \frac{1}{3}e^{-3i\theta}\right) &= 1 + \frac{1}{3}e^{3i\theta} + \frac{1}{3}e^{-3i\theta} + \frac{1}{9}e^{3i\theta}e^{-3i\theta} \\ &= 1 + \frac{1}{3}(e^{3i\theta} + e^{-3i\theta}) + \frac{1}{9} = \frac{10}{9} + \frac{2}{3}\cos 3\theta. \end{aligned}$$

12 b)

$$\begin{aligned} C + iS &= \cos\theta - \frac{1}{3}\cos 4\theta + \frac{1}{9}\cos 7\theta - \cdots + i(\sin\theta - \frac{1}{3}\sin 4\theta + \frac{1}{9}\sin 7\theta - \cdots) \\ &= (\cos\theta + i\sin\theta) - \frac{1}{3}(\cos 4\theta + i\sin 4\theta) + \frac{1}{9}(\cos 7\theta + i\sin 7\theta) - \cdots \\ &= e^{i\theta} - \frac{1}{3}e^{4i\theta} + \frac{1}{9}e^{7i\theta} - \cdots. \end{aligned}$$

$$a = e^{i\theta}, \quad r = -\frac{1}{3}e^{3i\theta}. \quad S_\infty = \frac{a}{1-r} = \frac{e^{i\theta}}{1 + \frac{1}{3}e^{3i\theta}}.$$

12 c)

$$\begin{aligned} C + iS &= \frac{e^{i\theta}}{1 + \frac{1}{3}e^{3i\theta}} \times \frac{1 + \frac{1}{3}e^{-3i\theta}}{1 + \frac{1}{3}e^{-3i\theta}} = \frac{e^{i\theta} + \frac{1}{3}e^{-2i\theta}}{\frac{10}{9} + \frac{2}{3}\cos 3\theta} \\ &= \frac{9e^{i\theta} + 3e^{-2i\theta}}{10 + 6\cos 3\theta} \\ &= \frac{9(\cos\theta + i\sin\theta) + 3(\cos(-2\theta) + i\sin(-2\theta))}{10 + 6\cos 3\theta} \\ &= \frac{9\cos\theta + 3\cos 2\theta + i(9\sin\theta - 3\sin 2\theta)}{10 + 6\cos 3\theta}. \end{aligned}$$

Compare reals:

$$C = \frac{9\cos\theta + 3\cos 2\theta}{10 + 6\cos 3\theta}.$$

12 d)

Compare imaginary:

$$S = \frac{9\sin\theta - 3\sin 2\theta}{10 + 6\cos 3\theta}.$$

13 a)

$$\begin{aligned} C+iS &= \cos 3\theta + \frac{1}{2}\cos 5\theta + \frac{1}{4}\cos 7\theta + \cdots + i(\sin 3\theta + \frac{1}{2}\sin 5\theta + \frac{1}{4}\sin 7\theta - \cdots) \\ &= (\cos 3\theta + i\sin 3\theta) + \frac{1}{2}(\cos 5\theta + i\sin 5\theta) + \frac{1}{4}(\cos 7\theta + i\sin 7\theta) - \cdots \\ &= e^{3i\theta} + \frac{1}{2}e^{5i\theta} + \frac{1}{4}e^{7i\theta} - \cdots . \end{aligned}$$

$$a = e^{3i\theta}, \quad r = \frac{1}{2}e^{2i\theta}, \quad S_\infty = \frac{e^{3i\theta}}{1-\frac{1}{2}e^{2i\theta}}.$$

$$\begin{aligned} C+iS &= \frac{e^{3i\theta}}{1-\frac{1}{2}e^{2i\theta}} \times \frac{1-\frac{1}{2}e^{-2i\theta}}{1-\frac{1}{2}e^{-2i\theta}} = \frac{e^{3i\theta}-\frac{1}{2}e^{i\theta}}{\frac{5}{4}-\frac{1}{2}(e^{2i\theta}+e^{-2i\theta})} \\ &= \frac{4e^{3i\theta}-2e^{i\theta}}{5-4\cos 2\theta} \\ &= \frac{4\cos 3\theta - 2\cos\theta + i(4\sin 3\theta - 2\sin\theta)}{5-4\cos 2\theta}. \end{aligned}$$

Compare reals:

$$C = \frac{4\cos 3\theta - 2\cos\theta}{5-4\cos 2\theta}.$$

13 b)

Compare imaginary:

$$S = \frac{4\sin 3\theta - 2\sin\theta}{5-4\cos 2\theta}.$$

14 a)

$$\begin{aligned} C+iS &= 3 + \cos\theta + \frac{1}{3}\cos 2\theta + \cdots + i(\sin\theta + \frac{1}{3}\sin 2\theta + \cdots) \\ &= 3 + (\cos\theta + i\sin\theta) + \frac{1}{3}(\cos 2\theta + i\sin 2\theta) + \cdots \\ &= 3 + e^{i\theta} + \frac{1}{3}e^{2i\theta} + \cdots . \end{aligned}$$

$$a = 3, \quad r = \frac{1}{3}e^{i\theta}, \quad S_\infty = \frac{3}{1-\frac{1}{3}e^{i\theta}}.$$

$$\begin{aligned} C+iS &= \frac{3}{1-\frac{1}{3}e^{i\theta}} \times \frac{1-\frac{1}{3}e^{-i\theta}}{1-\frac{1}{3}e^{-i\theta}} = \frac{3-e^{-i\theta}}{\frac{10}{9}-\frac{1}{3}(e^{i\theta}+e^{-i\theta})} \\ &= \frac{27-9e^{-i\theta}}{10-6\cos\theta} = \frac{27-9(\cos\theta - i\sin\theta)}{10-6\cos\theta} \\ &= \frac{27-9\cos\theta + i(9\sin\theta)}{10-6\cos\theta}. \end{aligned}$$

Compare reals:

$$C = \frac{27-9\cos\theta}{10-6\cos\theta}.$$

14 b)

Compare imaginary:

$$S = \frac{9\sin\theta}{10 - 6\cos\theta}.$$

15 a)

$$\left(1 - \frac{1}{2}e^{2i\theta}\right)\left(1 - \frac{1}{2}e^{-2i\theta}\right) = 1 - \frac{1}{2}e^{2i\theta} - \frac{1}{2}e^{-2i\theta} + \frac{1}{4} = \frac{5}{4} - \frac{1}{2}(e^{2i\theta} + e^{-2i\theta})$$
$$= \frac{5}{4} - \cos 2\theta \quad \text{as req.}$$

15 b)

$$C + iS = 2 + \cos 2\theta + \frac{1}{2}\cos 4\theta + \cdots + \frac{1}{512}\cos 18\theta + i(\sin 2\theta + \frac{1}{2}\sin 4\theta + \cdots + \frac{1}{512}\sin 18\theta)$$
$$= 2 + e^{2i\theta} + \frac{1}{2}e^{4i\theta} + \cdots + \frac{1}{256}e^{18i\theta}.$$

$$a = 2, \quad r = \frac{1}{2}e^{2i\theta}, \quad n = 10.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{2(1 - \frac{1}{2^{10}}e^{20i\theta})}{1 - \frac{1}{2}e^{2i\theta}} = \frac{2 - \frac{1}{512}e^{20i\theta}}{1 - \frac{1}{2}e^{2i\theta}}.$$

$$C + iS = \frac{2 - \frac{1}{512}e^{20i\theta}}{1 - \frac{1}{2}e^{2i\theta}} \times \frac{1 - \frac{1}{2}e^{-2i\theta}}{1 - \frac{1}{2}e^{-2i\theta}}$$
$$= \frac{2 - e^{-2i\theta} - \frac{1}{512}e^{20i\theta} + \frac{1}{1024}e^{18i\theta}}{\frac{5}{4} - \cos 2\theta}$$
$$= \frac{8 - 4e^{-2i\theta} - \frac{1}{128}e^{20i\theta} + \frac{1}{256}e^{18i\theta}}{5 - 4\cos 2\theta}$$
$$= \frac{2048 - 1024e^{-2i\theta} - 2e^{20i\theta} + e^{18i\theta}}{256(5 - 4\cos 2\theta)}$$
$$= \frac{2048 - 1024\cos 2\theta - 2\cos 20\theta + \cos 18\theta + i(1024\sin 2\theta - 2\sin 20\theta + \sin 18\theta)}{256(5 - 4\cos 2\theta)}.$$

Compare reals:

$$C = \frac{2048 - 1024\cos 2\theta - 2\cos 20\theta + \cos 18\theta}{256(5 - 4\cos 2\theta)} \quad \text{as req.}$$

15 c)

Compare imaginary:

$$S = \frac{1024\sin 2\theta - 2\sin 20\theta + \sin 18\theta}{256(5 - 4\cos 2\theta)} \quad \text{as req.}$$

16 a)

$$(1-e^{2i\theta})(1-e^{-2i\theta}) = 1 - e^{2i\theta} - e^{-2i\theta} + 1 = 2 - (e^{2i\theta} + e^{-2i\theta}) = 2 - 2\cos 2\theta.$$

16 b)

$$\begin{aligned} C + iS &= \cos\theta + \cos 3\theta + \cdots + \cos 45\theta + i(\sin\theta + \sin 3\theta + \cdots + \sin 45\theta) \\ &= e^{i\theta} + e^{3i\theta} + \cdots + e^{45i\theta}. \end{aligned}$$

$$a = e^{i\theta}, \quad r = e^{2i\theta}, \quad n = 23.$$

$$\begin{aligned} S_n &= \frac{e^{i\theta}(1-e^{46i\theta})}{1-e^{2i\theta}} = \frac{e^{i\theta} - e^{47i\theta}}{1-e^{2i\theta}}, \\ C + iS &= \frac{e^{i\theta} - e^{47i\theta}}{1-e^{2i\theta}} \times \frac{1-e^{-2i\theta}}{1-e^{-2i\theta}} = \frac{e^{i\theta} - e^{47i\theta} - e^{-i\theta} + e^{45i\theta}}{2 - (e^{2i\theta} + e^{-2i\theta})} \\ &= \frac{e^{i\theta} - e^{47i\theta} - e^{-i\theta} + e^{45i\theta}}{2 - 2\cos 2\theta} \\ &= \frac{(\cos\theta + i\sin\theta) - (\cos 47\theta + i\sin 47\theta) - (\cos(-\theta) + i\sin(-\theta)) + (\cos 45\theta + i\sin 45\theta)}{2 - 2\cos 2\theta} \\ &= \frac{\cos\theta - \cos 47\theta - \cos\theta + \cos 45\theta + i(\sin\theta - \sin 47\theta + \sin\theta + \sin 45\theta)}{2 - 2\cos 2\theta}. \end{aligned}$$

Compare reals:

$$C = \frac{-\cos 47\theta + \cos 45\theta}{2 - 2\cos 2\theta} \quad \text{as req.}$$

16 c)

Compare imaginary:

$$S = \frac{2\sin\theta - \sin 47\theta + \sin 45\theta}{2 - 2\cos 2\theta} \quad \text{as req.}$$

17 a)

$$\begin{aligned} \left(1 - \frac{1}{9}e^{3i\theta}\right)\left(1 - \frac{1}{9}e^{-3i\theta}\right) &= 1 - \frac{1}{9}e^{3i\theta} - \frac{1}{9}e^{-3i\theta} + \frac{1}{81} \\ &= 1 - \frac{1}{9}(e^{3i\theta} + e^{-3i\theta}) + \frac{1}{81} = \frac{82}{81} - \frac{9}{81}(2\cos 3\theta) \\ &= \frac{1}{81}(82 - 18\cos 3\theta) \quad \text{as req.} \end{aligned}$$

17 b)

$$a = e^{3i\theta}, \quad r = \frac{1}{9}e^{3i\theta}, \quad S_\infty = \frac{a}{1-r} = \frac{e^{3i\theta}}{1 - \frac{1}{9}e^{3i\theta}}.$$

17 c)

$$C + iS = \frac{e^{3i\theta}}{1 - \frac{1}{9}e^{3i\theta}} \times \frac{1 - \frac{1}{9}e^{-3i\theta}}{1 - \frac{1}{9}e^{-3i\theta}} = \frac{e^{3i\theta} - \frac{1}{9}}{\frac{1}{81}(82 - 18\cos 3\theta)}$$
$$= \frac{81\cos 3\theta + 81i\sin 3\theta - 9}{82 - 18\cos 3\theta}.$$

Compare reals:

$$C = \frac{81\cos 3\theta - 9}{82 - 18\cos 3\theta}.$$

17 d)

Compare imaginary:

$$S = \frac{81\sin 3\theta}{82 - 18\cos 3\theta}.$$

18)

$$C + iS = \frac{1}{3}\cos\theta + \frac{1}{9}\cos 2\theta + \cdots + \frac{1}{3^n}\cos n\theta + i\left(\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \cdots + \frac{1}{3^n}\sin n\theta\right)$$
$$= \frac{1}{3}(\cos\theta + i\sin\theta) + \frac{1}{9}(\cos 2\theta + i\sin 2\theta) + \cdots + \frac{1}{3^n}(\cos n\theta + i\sin n\theta)$$
$$= \frac{1}{3}e^{i\theta} + \frac{1}{9}e^{2i\theta} + \cdots + \frac{1}{3^n}e^{ni\theta}.$$

$$a = \frac{1}{3}e^{i\theta}, \quad r = \frac{1}{3}e^{i\theta}, \quad n = n.$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{\frac{1}{3}e^{i\theta}(1 - \frac{1}{3^n}e^{ni\theta})}{1 - \frac{1}{3}e^{i\theta}} = \frac{e^{i\theta}(1 - \frac{1}{3^n}e^{ni\theta})}{3 - e^{i\theta}}$$
$$= \frac{e^{i\theta} - \frac{1}{3^n}e^{(n+1)i\theta}}{3 - e^{i\theta}}.$$

$$C + iS = \frac{e^{i\theta} - \frac{1}{3^n}e^{(n+1)i\theta}}{3 - e^{i\theta}} \times \frac{3 - e^{-i\theta}}{3 - e^{-i\theta}}$$
$$= \frac{3e^{i\theta} - 1 - 3^{1-n}e^{(n+1)i\theta} + 3^{-n}e^{ni\theta}}{10 - 3(e^{i\theta} + e^{-i\theta})}$$
$$= \frac{3e^{i\theta} - 1 - 3^{1-n}e^{(n+1)i\theta} + 3^{-n}e^{ni\theta}}{10 - 6\cos\theta}$$
$$= \frac{3\cos\theta - 1 - 3^{1-n}\cos((n+1)\theta) + 3^{-n}\cos(n\theta) + i(3\sin\theta - 3^{1-n}\sin((n+1)\theta) + 3^{-n}\sin(n\theta))}{10 - 6\cos\theta}.$$

Compare reals:

$$C = \frac{3\cos\theta - 1 - 3^{1-n}\cos((n+1)\theta) + 3^{-n}\cos(n\theta)}{10 - 6\cos\theta}$$
$$= \frac{3^n(3\cos\theta - 1 - 3^{1-n}\cos((n+1)\theta) + 3^{-n}\cos(n\theta))}{3^n(10 - 6\cos\theta)}$$
$$= \frac{3^{n+1}\cos\theta - 3^n - 3\cos((n+1)\theta) + \cos(n\theta)}{3^n(10 - 6\cos\theta)} \quad \text{as req.}$$

19 a)

$$\begin{aligned} C + iS &= \cos\theta + \cos 3\theta + \cdots + \cos((2n-1)\theta) + i(\sin\theta + \sin 3\theta + \cdots + \sin((2n-1)\theta)) \\ &= e^{i\theta} + e^{3i\theta} + \cdots + e^{(2n-1)i\theta}. \end{aligned}$$

$$a = e^{i\theta}, \quad r = e^{2i\theta}, \quad n = n.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{e^{i\theta}(1-e^{2ni\theta})}{1-e^{2i\theta}} = \frac{e^{i\theta} - e^{(2n+1)i\theta}}{1-e^{2i\theta}}.$$

$$\begin{aligned} C + iS &= \frac{e^{i\theta} - e^{(2n+1)i\theta}}{1-e^{2i\theta}} \times \frac{1-e^{-2i\theta}}{1-e^{-2i\theta}} \\ &= \frac{e^{i\theta} - e^{(2n+1)i\theta} - e^{-i\theta} + e^{(2n-1)i\theta}}{2 - (e^{2i\theta} + e^{-2i\theta})} \\ &= \frac{(\cos\theta + i\sin\theta) - (\cos((2n+1)\theta) + i\sin((2n+1)\theta)) - (\cos(-\theta) + i\sin(-\theta)) + (\cos((2n-1)\theta) + i\sin((2n-1)\theta))}{2 - 2\cos 2\theta} \\ &= \frac{-\cos((2n+1)\theta) + \cos((2n-1)\theta) + i(2\sin\theta - \sin((2n+1)\theta) + \sin((2n-1)\theta))}{2 - 2\cos 2\theta}. \end{aligned}$$

Compare reals:

$$C = \frac{-\cos((2n+1)\theta) + \cos((2n-1)\theta)}{2 - 2\cos 2\theta} \quad \text{as req.}$$

19 b)

Compare imaginary:

$$S = \frac{2\sin\theta - \sin((2n+1)\theta) + \sin((2n-1)\theta)}{2 - 2\cos 2\theta} \quad \text{as req.}$$

20 a)

$$(1 + e^{i\theta})^4 = 1 + 4e^{i\theta} + 6e^{2i\theta} + 4e^{3i\theta} + e^{4i\theta}.$$

20 b)

$$(1 + e^{i\theta})^5 = 1 + 5e^{i\theta} + 10e^{2i\theta} + 10e^{3i\theta} + 5e^{4i\theta} + e^{5i\theta}.$$

20 c)

$$(1 + e^{i\theta})^n = 1 + \binom{n}{1}e^{i\theta} + \binom{n}{2}e^{2i\theta} + \cdots + \binom{n}{n}e^{ni\theta}.$$

20 d)

$$(1 + e^{2i\theta})^n = 1 + \binom{n}{1}e^{2i\theta} + \binom{n}{2}e^{4i\theta} + \cdots + \binom{n}{n}e^{2ni\theta}.$$

21 a)

$$1 + \cos 2A = 1 + (2\cos^2 A - 1) = 2\cos^2 A.$$

21 b)

$$\sin 2A = 2\sin A \cos A.$$

21 c)

$$1 + e^{i\theta} = 1 + \cos\theta + i\sin\theta.$$

Since $1 + \cos\theta = 2\cos^2(\frac{\theta}{2})$, then

$$1 + e^{i\theta} = 2\cos^2\left(\frac{\theta}{2}\right) + i\sin\theta.$$

Since $\sin\theta = 2\sin(\frac{\theta}{2})\cos(\frac{\theta}{2})$, then

$$\begin{aligned} 1 + e^{i\theta} &= 2\cos^2\left(\frac{\theta}{2}\right) + i\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) \\ &= 2\cos\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right)\right) \\ &= 2e^{i\theta/2}\cos\left(\frac{\theta}{2}\right) \quad \text{as req.} \end{aligned}$$

22 a)

RHS:

$$2e^{i\theta/2}\cos\left(\frac{\theta}{2}\right) = e^{i\theta/2}\left(2\cos\frac{\theta}{2}\right) = e^{i\theta/2}\left(e^{i\theta/2} + e^{-i\theta/2}\right) = e^{i\theta} + \frac{e^{i\theta}}{e^{i\theta}} = e^{i\theta} + 1 : \text{LHS} \ \therefore \ \text{True.}$$

22 b)

$$\begin{aligned} C + iS &= 1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta + i(4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta) \\ &= 1 + 4e^{i\theta} + 6e^{2i\theta} + 4e^{3i\theta} + e^{4i\theta} = (1 + e^{i\theta})^4. \end{aligned}$$

Using $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$\begin{aligned} C + iS &= \left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^4 = 16e^{4i\theta/2}\cos^4\frac{\theta}{2} \\ &= 16(\cos 2\theta + i\sin 2\theta)\cos^4\frac{\theta}{2}. \end{aligned}$$

Compare reals:

$$C = 16\cos 2\theta\cos^4\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

22 c)

Compare imaginary:

$$S = 16 \sin 2\theta \cos^4\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

23 a)

$$1 - e^{i\theta} = 1 - \cos\theta - i\sin\theta.$$

Since $1 - \cos\theta = 2\sin^2(\frac{\theta}{2})$, then

$$1 - e^{i\theta} = 2\sin^2\left(\frac{\theta}{2}\right) - i\sin\theta.$$

Since $\sin\theta = 2\sin(\frac{\theta}{2})\cos(\frac{\theta}{2})$, then

$$\begin{aligned}
1 - e^{i\theta} &= 2\sin^2\left(\frac{\theta}{2}\right) - i\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) \\
&= 2\sin\left(\frac{\theta}{2}\right)\left(\sin\left(\frac{\theta}{2}\right) - i\cos\left(\frac{\theta}{2}\right)\right) \\
&= 2i\sin\left(\frac{\theta}{2}\right)\left(-\cos\left(\frac{\theta}{2}\right) + \frac{1}{i}\sin\left(\frac{\theta}{2}\right)\right) \\
&= -2i\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right)\right) \\
&= -2ie^{i\theta/2}\sin\left(\frac{\theta}{2}\right) \quad \text{as req.}
\end{aligned}$$

Alternatively:

$$\begin{aligned}
\text{RHS: } -2ie^{i\theta/2}\sin\frac{\theta}{2} &= -e^{i\theta/2}\left(2i\sin\frac{\theta}{2}\right) = -e^{i\theta/2}\left(e^{i\theta/2} - e^{-i\theta/2}\right) \\
&= -e^{i\theta} + \frac{e^{i\theta/2}}{e^{i\theta/2}} = 1 - e^{i\theta}.
\end{aligned}$$

Which is the LHS $\therefore$ True.

23 b)

$$\begin{aligned}
C + iS &= 1 - 5\cos\theta + 10\cos 2\theta - \cdots - \cos 5\theta + i(-5\sin\theta + 10\sin 2\theta - \cdots - \sin 5\theta) \\
&= 1 - 5e^{i\theta} + 10e^{2i\theta} - 10e^{3i\theta} + 5e^{4i\theta} - e^{5i\theta} = (1 - e^{i\theta})^5.
\end{aligned}$$

Using $1 - e^{i\theta} = -2ie^{i\theta/2}\sin(\frac{\theta}{2})$, then

$$\begin{aligned}
C + iS &= \left(-2ie^{i\theta/2}\sin\frac{\theta}{2}\right)^5 = -32ie^{5i\theta/2}\sin^5\frac{\theta}{2} \\
&= -32i\left(\cos\frac{5\theta}{2} + i\sin\frac{5\theta}{2}\right)\sin^5\frac{\theta}{2} \\
&= \left(-32i\cos\frac{5\theta}{2} + 32\sin\frac{5\theta}{2}\right)\sin^5\frac{\theta}{2}.
\end{aligned}$$

Compare reals:

$$C = 32\sin\left(\frac{5\theta}{2}\right)\sin^5\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

23 c)

Compare imaginary:

$$S = -32\cos\left(\frac{5\theta}{2}\right)\sin^5\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

24 a)

$$\begin{aligned} C + iS &= 1 + 6\cos\theta + 15\cos 2\theta + \cdots + \cos 6\theta + i(6\sin\theta + 15\sin 2\theta + \cdots + \sin 6\theta) \\ &= 1 + 6e^{i\theta} + 15e^{2i\theta} + 20e^{3i\theta} + 15e^{4i\theta} + 6e^{5i\theta} + e^{6i\theta} = (1 + e^{i\theta})^6. \end{aligned}$$

Using $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$\begin{aligned} C + iS &= \left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^6 = 64e^{3i\theta}\cos^6\frac{\theta}{2} \\ &= 64(\cos 3\theta + i\sin 3\theta)\cos^6\frac{\theta}{2}. \end{aligned}$$

Compare reals:

$$C = 64\cos 3\theta\cos^6\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

24 b)

Compare imaginary:

$$S = 64\sin 3\theta\cos^6\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

25 a)

$$\begin{aligned} C + iS &= 1 + 6\cos 2\theta + 15\cos 4\theta + \cdots + \cos 12\theta + i(6\sin 2\theta + 15\sin 4\theta + \cdots + \sin 12\theta) \\ &= 1 + 6e^{2i\theta} + 15e^{4i\theta} + 20e^{6i\theta} + 15e^{8i\theta} + 6e^{10i\theta} + e^{12i\theta} = (1 + e^{2i\theta})^6. \end{aligned}$$

Using $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$\begin{aligned} C + iS &= (2e^{i\theta}\cos\theta)^6 = 64e^{6i\theta}\cos^6\theta \\ &= 64(\cos 6\theta + i\sin 6\theta)\cos^6\theta. \end{aligned}$$

Compare reals:

$$C = 64\cos 6\theta\cos^6\theta \quad \text{as req.}$$

25 b)

Compare imaginary:

$$S = 64\sin 6\theta\cos^6\theta \quad \text{as req.}$$

26 a)

$$\begin{aligned}C+iS &= 1+\binom{n}{1}\cos\theta+\binom{n}{2}\cos 2\theta+\cdots+\binom{n}{n}\cos n\theta+i\left(\binom{n}{1}\sin\theta+\binom{n}{2}\sin 2\theta+\cdots+\binom{n}{n}\sin n\theta\right)\\ &= 1+\binom{n}{1}(\cos\theta+i\sin\theta)+\binom{n}{2}(\cos 2\theta+i\sin 2\theta)+\cdots+\binom{n}{n}(\cos n\theta+i\sin n\theta)\\ &= 1+\binom{n}{1}e^{i\theta}+\binom{n}{2}e^{2i\theta}+\cdots+\binom{n}{n}e^{ni\theta}=(1+e^{i\theta})^n.\end{aligned}$$

Using $1+e^{i\theta}=2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$\begin{aligned}C+iS &= \left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^n = 2^n e^{in\theta/2}\cos^n\frac{\theta}{2}\\ &= 2^n\left(\cos\frac{n\theta}{2}+i\sin\frac{n\theta}{2}\right)\cos^n\frac{\theta}{2}.\end{aligned}$$

Compare reals:

$$C=2^n\cos\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)\quad\text{as req.}$$

26 b)

Compare imaginary:

$$S=2^n\sin\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)\quad\text{as req.}$$

27 a)

$$1+\cos 2A=1+(2\cos^2 A-1)=2\cos^2 A\quad\text{and}\quad\sin 2A=2\sin A\cos B.$$

Given that $1+e^{i\theta}=1+\cos\theta+i\sin\theta$, and since $1+\cos\theta=2\cos^2(\frac{\theta}{2})$, then

$$1+e^{i\theta}=2\cos^2\left(\frac{\theta}{2}\right)+i\sin\theta.$$

Also since $\sin\theta=2\sin(\frac{\theta}{2})\cos(\frac{\theta}{2})$, then

$$\begin{aligned}1+e^{i\theta} &= 2\cos^2\left(\frac{\theta}{2}\right)+i\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right)\\ &= 2\cos\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right)+i\sin\left(\frac{\theta}{2}\right)\right)\\ 1+e^{i\theta} &= 2e^{i\theta/2}\cos\left(\frac{\theta}{2}\right)\quad\text{as req.}\end{aligned}$$

27 b)

$$\begin{aligned}C+iS &= 1+3\cos\theta+3\cos 2\theta+\cos 3\theta+i(3\sin\theta+3\sin 2\theta+\sin 3\theta)\\ &= 1+3e^{i\theta}+3e^{2i\theta}+e^{3i\theta}=(1+e^{i\theta})^3.\end{aligned}$$

Using $1+e^{i\theta}=2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$C+iS=\left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^3=8e^{3i\theta/2}\cos^3\frac{\theta}{2}$$

$$= 8\left(\cos\frac{3\theta}{2} + i\sin\frac{3\theta}{2}\right)\cos^3\frac{\theta}{2}.$$

Compare reals:

$$C = 8\cos\left(\frac{3\theta}{2}\right)\cos^3\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

Compare imaginary:

$$S = 8\sin\left(\frac{3\theta}{2}\right)\cos^3\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

28 a)

$$a = 1, \quad r = z, \quad S_\infty = \frac{a}{1-r} = \frac{1}{1-z}.$$

28 b)

If $z = \frac{1}{3}(\cos\theta + i\sin\theta)$, then $z = \frac{1}{3}e^{i\theta}$.

$$\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots \text{ is the imaginary part of } \sum_{n=1}^{\infty}\frac{1}{3^n}e^{ni\theta}.$$

$$a = \frac{1}{3}e^{i\theta}, \quad r = \frac{1}{3}e^{i\theta}, \quad S_\infty = \frac{a}{1-r} = \frac{\frac{1}{3}e^{i\theta}}{1-\frac{1}{3}e^{i\theta}} = \frac{e^{i\theta}}{3-e^{i\theta}}.$$

$$\begin{aligned} S_\infty &= \frac{e^{i\theta}}{3-e^{i\theta}} \times \frac{3-e^{-i\theta}}{3-e^{-i\theta}} = \frac{3e^{i\theta}-1}{9-3(e^{i\theta}+e^{-i\theta})+1} \\ &= \frac{3(\cos\theta + i\sin\theta)-1}{10-6\cos\theta} = \frac{3\cos\theta - 1 + i(3\sin\theta)}{10-6\cos\theta}. \end{aligned}$$

Compare imaginary:

$$\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots = \frac{3\sin\theta}{10-6\cos\theta}.$$

28 c)

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}.$$

For a purely imaginary number, $\mathrm{Re}\left(\frac{1}{1-z}\right) = 0$. But $|z| < 1$ and so $\mathrm{Re}(1-z) > 0$, $\therefore \mathrm{Re}\left(\frac{1}{1-z}\right) \neq 0$.

29 a)

$$a = z, \quad r = z, \quad S_\infty = \frac{a}{1-r} = \frac{z}{1-z}.$$

29 b)

If $z = \frac{1}{4}(\cos\theta + i\sin\theta)$, then $z = \frac{1}{4}e^{i\theta}$.

$$\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots \text{ is the real part of } \sum_{n=1}^{\infty} \frac{1}{4^n}e^{ni\theta}.$$

$$a = \frac{1}{4}e^{i\theta}, \quad r = \frac{1}{4}e^{i\theta}.$$

$$\begin{aligned} S_\infty &= \frac{a}{1-r} = \frac{\frac{1}{4}e^{i\theta}}{1-\frac{1}{4}e^{i\theta}} = \frac{e^{i\theta}}{4-e^{i\theta}} \\ &= \frac{e^{i\theta}}{4-e^{i\theta}} \times \frac{4-e^{-i\theta}}{4-e^{-i\theta}} = \frac{4e^{i\theta}-1}{16-4(e^{i\theta}+e^{-i\theta})+1} \\ &= \frac{4(\cos\theta + i\sin\theta)-1}{17-8\cos\theta} = \frac{4\cos\theta - 1 + i(4\sin\theta)}{17-8\cos\theta}. \end{aligned}$$

Compare real:

$$\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots = \frac{4\cos\theta - 1}{17-8\cos\theta}.$$

29 c)

Imaginary part:

$$\frac{4\sin\theta}{17-8\cos\theta}.$$

This must be zero for it to be purely real, so $4\sin\theta = 0$.

$$\therefore \theta = k\pi \quad \text{where } k \in \mathbb{Z}.$$

30 a)

Since $z = \frac{1}{2}e^{i\theta}$:

$$\frac{|z^{n+1}|}{|z^n|} = \left|\frac{\frac{1}{2^{n+1}}e^{(n+1)i\theta}}{\frac{1}{2^n}e^{ni\theta}}\right| = \left|\frac{1}{2}\frac{e^{ni\theta}e^{i\theta}}{e^{ni\theta}}\right| = \frac{1}{2}|e^{i\theta}| = \frac{1}{2}.$$

30 b)

$$\theta.$$

30 c)

$$\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots \text{ is the imaginary part of } \sum_{n=1}^{\infty} \frac{1}{2^n}e^{ni\theta}.$$

$$a = \frac{1}{2}e^{i\theta}, \quad r = \frac{1}{2}e^{i\theta}.$$

$$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}e^{i\theta}}{1-\frac{1}{2}e^{i\theta}} = \frac{e^{i\theta}}{2-e^{i\theta}}$$

$$= \frac{e^{i\theta}}{2-e^{i\theta}} \times \frac{2-e^{-i\theta}}{2-e^{-i\theta}} = \frac{2e^{i\theta}-1}{4-2(e^{i\theta}+e^{-i\theta})+1}$$
$$= \frac{2(\cos\theta + i\sin\theta)-1}{5-4\cos\theta} = \frac{2\cos\theta - 1 + i(2\sin\theta)}{5-4\cos\theta}.$$

Compare imaginary:

$$\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots = \frac{2\sin\theta}{5-4\cos\theta} \quad \text{as req.}$$

31)

$$C + iS = \frac{1}{2}\cos\theta + \frac{1}{4}\cos 2\theta + \cdots + \frac{1}{2^n}\cos n\theta + i\left(\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \cdots + \frac{1}{2^n}\sin n\theta\right)$$
$$= \frac{1}{2}e^{i\theta} + \frac{1}{4}e^{2i\theta} + \cdots + \frac{1}{2^n}e^{ni\theta}.$$
$$a = \frac{1}{2}e^{i\theta}, \quad r = \frac{1}{2}e^{i\theta}, \quad n = n.$$
$$S_n = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{2}e^{i\theta}(1-\frac{1}{2^n}e^{ni\theta})}{1-\frac{1}{2}e^{i\theta}} = \frac{e^{i\theta}-\frac{1}{2^n}e^{(n+1)i\theta}}{2-e^{i\theta}}.$$

$$C + iS = \frac{e^{i\theta}-\frac{1}{2^n}e^{(n+1)i\theta}}{2-e^{i\theta}} \times \frac{2-e^{-i\theta}}{2-e^{-i\theta}}$$
$$= \frac{2e^{i\theta} - 1 - 2^{1-n}e^{(n+1)i\theta} + 2^{-n}e^{ni\theta}}{5-2(e^{i\theta}+e^{-i\theta})}$$
$$= \frac{2\cos\theta - 1 - 2^{1-n}\cos((n+1)\theta) + 2^{-n}\cos(n\theta) + i(2\sin\theta - 2^{1-n}\sin((n+1)\theta) + 2^{-n}\sin(n\theta))}{5-4\cos\theta}.$$

Compare imaginary:

$$S = \frac{2\sin\theta - 2^{1-n}\sin((n+1)\theta) + 2^{-n}\sin(n\theta)}{5-4\cos\theta}$$
$$= \frac{2^n(2\sin\theta - 2^{1-n}\sin((n+1)\theta) + 2^{-n}\sin(n\theta))}{2^n(5-4\cos\theta)}$$
$$= \frac{2^{n+1}\sin\theta - 2\sin((n+1)\theta) + \sin(n\theta)}{2^n(5-4\cos\theta)} \quad \text{as req.}$$

32)

$$C + iS = 1 + \binom{12}{1}\cos\theta + \cdots + \cos 12\theta + i\left(1 + \binom{12}{1}\sin\theta + \cdots + \sin 12\theta\right)$$
$$= 1 + \binom{12}{1}e^{i\theta} + \cdots + e^{12i\theta} = (1+e^{i\theta})^{12}.$$

Using $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$C + iS = \left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^{12} = 4096e^{6i\theta}\cos^{12}\frac{\theta}{2}$$
$$= 4096(\cos 6\theta + i\sin 6\theta)\cos^{12}\frac{\theta}{2}.$$

Compare imaginary:

$$\therefore S = 4096\sin 6\theta\cos^{12}\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

33)

$$C + iS = 1 - \binom{16}{1}\cos\theta + \cdots + \cos 16\theta + i\left(1 - \binom{16}{1}\sin\theta + \cdots + \sin 16\theta\right)$$
$$= 1 - \binom{16}{1}e^{i\theta} + \cdots + e^{16i\theta} = (1 - e^{i\theta})^{16}.$$

Using $1 - e^{i\theta} = -2ie^{i\theta/2}\sin(\frac{\theta}{2})$, then

$$C + iS = \left(-2ie^{i\theta/2}\sin\frac{\theta}{2}\right)^{16} = 2^{16}e^{8i\theta}\sin^{16}\frac{\theta}{2}$$
$$= 2^{16}(\cos 8\theta + i\sin 8\theta)\sin^{16}\frac{\theta}{2}.$$

Compare real:

$$\therefore C = 2^{16}\cos 8\theta \sin^{16}\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

34 a)

$$\frac{2}{1 - e^{2i\theta}} \times \frac{1 - e^{-2i\theta}}{1 - e^{-2i\theta}} = \frac{2 - 2e^{-2i\theta}}{1 - e^{2i\theta} - e^{-2i\theta} + 1} = \frac{2 - 2e^{-2i\theta}}{2 - (e^{2i\theta} + e^{-2i\theta})}$$
$$= \frac{1 - e^{-2i\theta}}{1 - \cos 2\theta}.$$

Since $e^{-2i\theta} = \cos(-2\theta) + i\sin(-2\theta) = \cos 2\theta - i\sin 2\theta$, then

$$1 - e^{-2i\theta} = 1 - \cos 2\theta + i\sin 2\theta.$$

So

$$\frac{1 - e^{-2i\theta}}{1 - \cos 2\theta} = 1 + \frac{\sin 2\theta}{1 - \cos 2\theta}i = 1 + \frac{2\cos\theta\sin\theta}{2\sin^2\theta}i$$
$$= 1 + \frac{\cos\theta}{\sin\theta}i = 1 + i\cot\theta \quad \text{as req.}$$

34 b)

$$C + iS = 2 + 2\cos\frac{\pi}{14} + 2\cos\frac{2\pi}{14} + \cdots + 2\cos\frac{6\pi}{14} + 2i\left(\sin\frac{\pi}{14} + \cdots + \sin\frac{6\pi}{14}\right)$$
$$= 2e^0 + 2e^{i\pi/14} + 2e^{2i\pi/14} + \cdots + 2e^{6i\pi/14}.$$
$$a = 2e^0, \quad r = e^{i\pi/14}, \quad n = 7.$$
$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{2(1 - e^{i\pi/2})}{1 - e^{i\pi/14}}.$$

But since $\frac{2}{1-e^{2i\theta}} = 1 + i\cot\theta$, set $\theta = \frac{\pi}{28}$:

$$C + iS = \frac{2(1 - e^{i\pi/2})}{1 - e^{i\pi/14}} = \left(1 + i\cot\frac{\pi}{28}\right)(1 - e^{i\pi/2})$$
$$= \left(1 + i\cot\frac{\pi}{28}\right)(1 - i) = 1 + i\cot\frac{\pi}{28} - i + \cot\frac{\pi}{28}.$$

So that

$$C = 1 + \cot\frac{\pi}{28}.$$

34 c)

$$S = -1 + \cot\frac{\pi}{28}.$$

And so

$$C - 2 = 1 + \cot\frac{\pi}{28} - 2 = -1 + \cot\frac{\pi}{28} = S, \quad \therefore S = C - 2.$$

35 a)

$$\begin{aligned} C + iS &= e^{i\theta} + e^{i(\theta+4\pi/5)} + e^{i(\theta+8\pi/5)} + e^{i(\theta+12\pi/5)} + e^{i(\theta+16\pi/5)} \\ &= e^{i\theta}(1 + e^{4\pi i/5} + e^{8\pi i/5} + e^{12\pi i/5} + e^{16\pi i/5}) \\ &= e^{i\theta}\sum_{k=0}^{4}(e^{4\pi i/5})^k \quad \text{as req.} \end{aligned}$$

35 b)

$1 + e^{4\pi i/5} + e^{8\pi i/5} + e^{12\pi i/5} + e^{16\pi i/5}$ is a finite geometric series.

$$a = 1, \quad r = e^{4\pi i/5}, \quad n = 5.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1-e^{4\pi i}}{1-e^{4\pi i/5}} = \frac{1-1}{1-e^{4\pi i/5}} = 0.$$

$$\therefore C + iS = e^{i\theta} \times 0.$$

And so $C = 0$ and $S = 0$.

36 a)

$$C + iS = e^{0i} - e^{\pi i/2} + e^{\pi i} - e^{3\pi i/2} = \sum_{k=0}^{3}(-1)^k(e^{\pi i/2})^k.$$

36 b)

$e^{0i} - e^{\pi i/2} + e^{\pi i} - e^{3\pi i/2}$ is a finite geometric series.

$$a = 1, \quad r = -e^{\pi i/2}, \quad n = 4.$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1-e^{2\pi i}}{1+e^{\pi i/2}} = \frac{1-1}{1+e^{\pi i/2}} = 0.$$

$$\therefore C + iS = 0.$$

And so $C = 0$ and $S = 0$.

37 a)

$$\begin{aligned} C + iS &= 1 + k(\cos\theta + i\sin\theta) + k^2(\cos 2\theta + i\sin 2\theta) + \cdots \\ &= 1 + ke^{i\theta} + k^2e^{2i\theta} + \cdots . \end{aligned}$$

$$a = 1, \quad r = ke^{i\theta}, \quad S_\infty = \frac{1}{1 - ke^{i\theta}}.$$

$$\begin{aligned} C + iS &= \frac{1}{1 - ke^{i\theta}} \times \frac{1 - ke^{-i\theta}}{1 - ke^{-i\theta}} = \frac{1 - ke^{-i\theta}}{1 + k^2 - k(e^{i\theta} + e^{-i\theta})} \\ &= \frac{1 - k(\cos(-\theta) + i\sin(-\theta))}{1 + k^2 - 2k\cos\theta} = \frac{1 - k\cos\theta + ki\sin\theta}{1 - 2k\cos\theta + k^2}. \end{aligned}$$

Compare reals:

$$C = \frac{1 - k\cos\theta}{1 - 2k\cos\theta + k^2} \quad \text{as req.}$$

37 b)

Compare imaginary:

$$S = \frac{k\sin\theta}{1 - 2k\cos\theta + k^2}.$$

38 a)

$$\begin{aligned} C + iS &= ke^{i\theta} + k^2e^{2i\theta} + k^3e^{3i\theta} + \cdots , \\ a &= ke^{i\theta}, \quad r = ke^{i\theta}, \quad S_\infty = \frac{ke^{i\theta}}{1 - ke^{i\theta}}. \end{aligned}$$

$$\begin{aligned} C + iS &= \frac{ke^{i\theta}}{1 - ke^{i\theta}} \times \frac{1 - ke^{-i\theta}}{1 - ke^{-i\theta}} = \frac{ke^{i\theta} - k^2}{1 + k^2 - k(e^{i\theta} + e^{-i\theta})} \\ &= \frac{k(\cos\theta + i\sin\theta) - k^2}{1 - 2k\cos\theta + k^2}. \end{aligned}$$

Compare reals:

$$C = \frac{k\cos\theta - k^2}{1 - 2k\cos\theta + k^2} \quad \text{as req.}$$

38 b)

Compare imaginary:

$$S = \frac{k\sin\theta}{1 - 2k\cos\theta + k^2}.$$

39 a)

$$\begin{aligned} C + iS &= 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \binom{n}{n}\cos n\theta + i\left(\binom{n}{1}\sin\theta + \cdots + \binom{n}{n}\sin n\theta\right) \\ &= 1 + \binom{n}{1}(\cos\theta + i\sin\theta) + \cdots + \binom{n}{n}(\cos n\theta + i\sin n\theta) \end{aligned}$$

$$= 1 + \binom{n}{1} e^{i\theta} + \cdots + \binom{n}{n} e^{ni\theta} = (1 + e^{i\theta})^n.$$

Using $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\frac{\theta}{2})$, then

$$\begin{aligned} C + iS &= \left(2e^{i\theta/2}\cos\frac{\theta}{2}\right)^n = 2^n e^{in\theta/2}\cos^n\frac{\theta}{2} \\ &= 2^n\left(\cos\frac{n\theta}{2} + i\sin\frac{n\theta}{2}\right)\cos^n\frac{\theta}{2}. \end{aligned}$$

Compare reals:

$$C = 2^n\cos\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right) \quad \text{as req.}$$

39 b)

Compare imaginary:

$$S = 2^n\sin\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right).$$

39 c)

$$\frac{C}{S} = \frac{2^n\cos(\frac{n\theta}{2})\cos^n(\frac{\theta}{2})}{2^n\sin(\frac{n\theta}{2})\cos^n(\frac{\theta}{2})} = \frac{\cos(\frac{n\theta}{2})}{\sin(\frac{n\theta}{2})} = \cot\left(\frac{n\theta}{2}\right).$$

40 a)

$$\begin{aligned} 1 - e^{i\theta} &= 1 - \frac{e^{i\theta}}{e^{i\theta/2}} = e^{i\theta/2}(e^{-i\theta/2} - e^{i\theta/2}) = -e^{i\theta/2}(e^{i\theta/2} - e^{-i\theta/2}) \\ &= -e^{i\theta/2}\left(2i\sin\frac{\theta}{2}\right) = -2ie^{i\theta/2}\sin\left(\frac{\theta}{2}\right) \quad \text{as req.} \end{aligned}$$

40 b)

$$\begin{aligned} C + iS &= 1 - \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + (-1)^n\binom{n}{n}\cos n\theta \\ &\quad + i\left(-\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + (-1)^n\binom{n}{n}\sin n\theta\right) \\ &= 1 - \binom{n}{1}e^{i\theta} + \cdots + (-1)^n\binom{n}{n}e^{ni\theta} = (1 - e^{i\theta})^n. \end{aligned}$$

Using $1 - e^{i\theta} = -2ie^{i\theta/2}\sin\frac{\theta}{2}$, then

$$C + iS = \left(-2ie^{i\theta/2}\sin\frac{\theta}{2}\right)^n = \left[-2i\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)\sin\frac{\theta}{2}\right]^n \quad \text{as req.}$$

40 c)

Let $n = 2k$ so that n is even:

$$\begin{aligned} C + iS &= \left[-2i\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)\sin\frac{\theta}{2}\right]^{2k} \\ &= (-2)^{2k}(i^{2k})\left(\cos\frac{2k\theta}{2} + i\sin\frac{2k\theta}{2}\right)\sin^{2k}\frac{\theta}{2}. \end{aligned}$$

Compare reals: $C = (-2)^{2k}(i^{2k})\cos k\theta \sin^{2k}\frac{\theta}{2}$. Compare imaginary: $S = (-2)^{2k}(i^{2k})\sin k\theta \sin^{2k}\frac{\theta}{2}$. So

$$\frac{S}{C} = \frac{(-2)^{2k}(i^{2k})\sin k\theta \sin^{2k}\frac{\theta}{2}}{(-2)^{2k}(i^{2k})\cos k\theta \sin^{2k}\frac{\theta}{2}} = \frac{\sin k\theta}{\cos k\theta}.$$

But $k = \frac{n}{2}$ and so

$$\frac{S}{C} = \frac{\sin(\frac{n}{2}\theta)}{\cos(\frac{n}{2}\theta)} = \tan\left(\frac{n}{2}\theta\right) \quad \text{as req.}$$

41 a)

$$\begin{aligned} C + iS &= \int_0^{\pi/3} e^{3x}\cos 5x\,dx + i\int_0^{\pi/3} e^{3x}\sin 5x\,dx \\ &= \int_0^{\pi/3} (e^{3x}\cos 5x + ie^{3x}\sin 5x)\,dx = \int_0^{\pi/3} e^{3x}(\cos 5x + i\sin 5x)\,dx \\ &= \int_0^{\pi/3} e^{3x}e^{5ix}\,dx = \int_0^{\pi/3} e^{(3+5i)x}\,dx \\ &= \left[\frac{1}{3+5i}e^{(3+5i)x}\right]_0^{\pi/3} = \frac{e^{\frac{\pi}{3}(3+5i)} - 1}{3+5i}. \end{aligned}$$

Since

$$e^{\frac{\pi}{3}(3+5i)} = e^{\pi}e^{5\pi i/3} = e^{\pi}\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right) = e^{\pi}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right),$$

then

$$\begin{aligned} C + iS &= \frac{e^{\pi}(\frac{1}{2} - \frac{\sqrt{3}}{2}i) - 1}{3+5i} \times \frac{3-5i}{3-5i} \\ &= \frac{\frac{3}{2}e^{\pi} - \frac{3\sqrt{3}}{2}ie^{\pi} - 3 - 5i}{34} \\ &= \frac{3e^{\pi} - 6 - 5\sqrt{3}e^{\pi} + (10 - 3\sqrt{3}e^{\pi} - 5e^{\pi})i}{68}. \end{aligned}$$

Compare reals:

$$C = \frac{3e^{\pi} - 6 - 5\sqrt{3}e^{\pi}}{68} = \frac{e^{\pi}}{68}(3 - 5\sqrt{3}) - \frac{3}{34} \quad \text{as req.}$$

41 b)

Compare imaginary:

$$S = \frac{10 - 3\sqrt{3}e^{\pi} - 5e^{\pi}}{68} = -\frac{e^{\pi}}{68}(5 + 3\sqrt{3}) + \frac{5}{34}.$$

42 a)

$$\begin{aligned} C + iS &= 1 + \cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta + i(\sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta) \\ &= 1 + e^{i\theta} + e^{2i\theta} + e^{3i\theta} + e^{4i\theta}. \end{aligned}$$

This is a finite geometric series with $a = 1, r = e^{i\theta}, n = 5$, so

$$S_n = \frac{1 - e^{5i\theta}}{1 - e^{i\theta}}.$$

$$\therefore C + iS = \frac{1 - e^{5i\theta}}{1 - e^{i\theta}} = \frac{e^{5i\theta/2} - e^{-5i\theta/2}}{e^{i\theta/2} - e^{-i\theta/2}} \times e^{2i\theta} \quad \text{as req.}$$

42 b)

Since $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$,

$$\frac{2i\sin(\frac{5}{2}\theta)}{2i\sin(\frac{1}{2}\theta)} \times e^{2i\theta} = \frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)}(\cos 2\theta + i\sin 2\theta).$$

Real part:

$$C = \cos 2\theta \times \frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)}.$$

Since $\sin(\frac{5}{2}\theta) = \sin(2\theta + \frac{1}{2}\theta) = \sin 2\theta\cos\frac{\theta}{2} + \cos 2\theta\sin\frac{\theta}{2}$, then

$$\begin{aligned} C &= \cos 2\theta \times \frac{\sin 2\theta\cos(\frac{1}{2}\theta) + \cos 2\theta\sin(\frac{1}{2}\theta)}{\sin(\frac{1}{2}\theta)} \\ &= \frac{\cos 2\theta\sin 2\theta\cos(\frac{1}{2}\theta)}{\sin(\frac{1}{2}\theta)} + \frac{\cos^2(2\theta)\sin(\frac{1}{2}\theta)}{\sin(\frac{1}{2}\theta)} \\ &= \frac{1}{2}\sin 4\theta\cot\left(\frac{\theta}{2}\right) + \cos^2(2\theta). \end{aligned}$$

42 c)

Since $C = \cos(2\theta)\frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)}$ and $S = \sin(2\theta)\frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)}$, then either $\frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)} = 0$ or $\sin 2\theta = \cos 2\theta$. The solutions for $\frac{\sin(\frac{5}{2}\theta)}{\sin(\frac{1}{2}\theta)} = 0$ are when $\sin(\frac{5}{2}\theta) = 0$, so $\theta = \frac{2\pi}{5}$ or $\frac{4\pi}{5}$. The solutions for $\sin 2\theta = \cos 2\theta$ are when $\tan 2\theta = 1$, so $\theta = \frac{\pi}{8}$ or $\frac{5\pi}{8}$.

$$\therefore \frac{2\pi}{5}, \quad \frac{4\pi}{5}, \quad \frac{\pi}{8} \quad \text{or} \quad \frac{5\pi}{8}.$$

43 a)

$$P = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + e^{4i\theta}.$$

$$P\big|_{\theta=2n\pi} = e^{2ni\pi} + e^{4ni\pi} + e^{6ni\pi} + e^{8ni\pi} = (e^{2i\pi})^n + (e^{2i\pi})^{2n} + (e^{2i\pi})^{3n} + (e^{2i\pi})^{4n} = 1 + 1 + 1 + 1 = 4.$$

43 b)

$$P = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + e^{4i\theta} \text{ is a finite geometric series.}$$

$$a = e^{i\theta}, \quad r = e^{i\theta}, \quad n = 4, \quad S_n = \frac{a(1-r^n)}{1-r} = \frac{e^{i\theta}(1-e^{4i\theta})}{1-e^{i\theta}}.$$

$$\therefore P = \frac{e^{i\theta}(1-e^{4i\theta})}{1-e^{i\theta}} = \frac{e^{i\theta/2}}{e^{i\theta/2}} \times \frac{1-e^{4i\theta}}{e^{-i\theta/2} - e^{i\theta/2}} = e^{i\theta/2} \times \frac{e^{4i\theta}-1}{e^{i\theta/2} - e^{-i\theta/2}}.$$

Since $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$, then

$$P = e^{i\theta/2}\frac{e^{4i\theta}-1}{2i\sin(\frac{\theta}{2})} \quad \text{as req.}$$

43 c)

$$\begin{aligned} P &= \frac{e^{i\theta/2}(e^{4i\theta}-1)}{2i\sin(\frac{\theta}{2})} \times \frac{-i}{-i} = \frac{-ie^{9i\theta/2} + ie^{i\theta/2}}{2\sin(\frac{\theta}{2})} \\ &= \frac{-i(\cos\frac{9\theta}{2} + i\sin\frac{9\theta}{2}) + i(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})}{2\sin(\frac{\theta}{2})} \\ &= \frac{-i\cos\frac{9\theta}{2} + \sin\frac{9\theta}{2} + i\cos\frac{\theta}{2} - \sin\frac{\theta}{2}}{2\sin(\frac{\theta}{2})}. \end{aligned}$$

$$\mathrm{Re}(P) = \frac{\sin(\frac{9}{2}\theta) - \sin(\frac{1}{2}\theta)}{2\sin(\frac{1}{2}\theta)} = \frac{\sin(\frac{9}{2}\theta)}{2\sin(\frac{1}{2}\theta)} - \frac{1}{2} \quad \text{as req.}$$

43 d)

$$\frac{\sin(\frac{9}{2}\theta)}{2\sin(\frac{1}{2}\theta)} - \frac{1}{2} = 0 \Rightarrow \frac{\sin(\frac{9}{2}\theta)}{2\sin(\frac{1}{2}\theta)} = \frac{1}{2} \Rightarrow \sin\left(\frac{9}{2}\theta\right) = \sin\left(\frac{1}{2}\theta\right).$$

Since $\sin x = \sin(\pi - x)$,

$$\frac{9}{2}\theta = \pi - \frac{1}{2}\theta \Rightarrow 5\theta = \pi \quad \therefore \theta = \frac{\pi}{5} \quad \text{as req.}$$

Since $\sin x = \sin(x + 2\pi)$,

$$\frac{9}{2}\theta = \frac{1}{2}\theta + 2\pi \Rightarrow 4\theta = 2\pi \quad \therefore \theta = \frac{\pi}{2}.$$